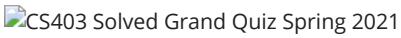
CS403 Solved Grand Quiz Spring 2021

June 22, 2021June 20, 2021 by Muhammad Junaid

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CS403 SOLVED MCQs

Which is not the function of Database Management System?

- A user accessible Catalog
- Concurrency Control Services
- **Failure Response Service**
- Support for Data Communication

The conceptual database design can be transformed into any _____.

- **Data model**
- Database design
- E-R model
- Entity set

Which is the second stage in the database development process?

- Database design
- Preliminary Study
- Physical Design
- **Requirement Analysis**

Functional dependencies under interference rule reflexivity are called _____ dependencies.

- Augmentation
- Transitivity
- Projectivity
- **Trivial**

If many instances of the “Book” entity are associated with the many instances of the “Student” entity, then the relationship between them is:

- One to one

- **Many to many**
- Many to one

Which of the following constraints enforces entity integrity?

a. **PRIMARY KEY**

- FOREIGN KEY
- CHECK
- NOT NULL

Unary operations involves

- **Only one relation**
- Only two relations
- More than two relations
- Not more than five relations

A _____ entity has a primary key that is partially or totally derived from the parent entity in the relationship.

- Strong
- **Weak**

- o Business
- Relationship

In _____ appropriate data model is chosen.

- ERD
- Conceptual Database Design
- **Logical Database Design**
- DFA

Which of the following is drawback of standardization?

- Difficult to understand
- Uniformity
- **Lack of uniqueness**
- Easy to develop applications

Recursive relationship is also called _____.

- **Unary**
- Ternary
- Primary
- Secondary

The foreign key attribute, which is present as a primary key in another relation is called as

_____ of the foreign key attribute.

- Binary relation

- **Home relation**
- Recursive relation

DFD does not provide us a way to represent _____

- Entities
- **Decisions**
- Date stores
- Processes

A _____ is used to maintain a connection between the users of the database system.

- Virtual server
- **File server**
- Client server
- FTP server

Which of the following is an advantage of using the Traditional File Organization?

- Time consuming
- Data security
- **Simplicity**
- Efficiency

Which of the following is the input to Analysis Phase?

- **DFD**
- Logical database design
- Physical database design
- Data model

Name and _____ are part of the definitions of an attribute.

a. a. Domain

- Symbol
- Property
- Entity instance

Which of the following most certainly implies the need for an entire table to implement?

- **A ternary relationship**
- A recursive relationship
- An identify relationship
- A binary relationship

Core of the Database Architecture is _____

- **Internal level**
- External level
- Conceptual level
- ANSI SPARK

- **Natural join**
- Semi join
- Outer join
- Theta join

Conceptual database design is implemented using a _____.

- Hierarchical data model
- **Semantic data model**
- Relational data model
- Network data model

Which constraint is generally used for efficiency purposes in the data entry process?

- Null value
- Integer value
- **Default value**
- String value

A 3 NF relation is converted to BCNF by

- Removing composite keys
- Removing multivalued dependencies
- **Dependent attributes of overlapping composite keys are put in a separate relation**
- Dependent non-key attributes are put in a separate table

Each _____ of a table contains atomic/single value.

- **Cell**
- Row
- Column
- Record

A relation is said to be in BCNF when

- It has overlapping composite keys
- It has no composite keys
- It has no multivalued dependencies
- **It has no overlapping composite keys which have related attributes**

Which one of the following is not a part of three-level architecture?

- External level
- Conceptual level
- Internal level
- **Physical level**

Which of the following is INCORRECT about naming entities?

- Singular nouns are used
- Write in capitals

- **Always use abbreviations as they are convenient**

An instance is

- A particular occurance of an entity
- A special type of relation
- An attribute of an entity
- **Any particular entity**

_____ is an example of metadata

- Ali
- Address
- Person
- **Student**

Level of data at which entities or objects exist in reality is called

- **Real world data**
- Schema for real world data
- Meta data
- Data about data

There are _____ basic properties of the database relations.

- Four
- Three
- **Six**
- Two

How many types of dependencies are?

- One
- Two
- **Three**
- Four

In _____ join, only selected rows of a relation are made cross product with second relation.

- Normal
- **Theta**
- Outer
- Semi

Which of the following is not a key?

- Super key
- Alternate key
- **Integrity constraint key**
- o Primary key

A collection of concepts that can be used to describe the structure of a database

- DBMS
- **Data model**
- Data

Any combination of attributes with super key is ____

- Primary key
- Foreign key
- **Super key**
- Candidate key

Binary relationships are those, which are established between

- One entity type
- **Two entity type**
- Three entity type
- Unlimited entities

_____ are called participants, when enrolled in a relationship.

- Attributes
- Relations
- **Entities**
- Keys

Incase of Context-level Diagram, the system is represented by

- One process atleast
- Two processes atleast
- **One process only**
- Any number of processes

Collection of raw facts is called _____.

- Information
- Object
- Output
- **Data**

A software package designed to store and manage database

- Database
- **DBMS**
- Data model
- Data

In _____ join common attributes appear twice in the output with selected rows.

- Normal join
- Semi join
- Natural join

Within a table, each primary key value _____.

- Is a minimal super key
- Is always the first field in each table
  - o Must be numeric
- **Must be unique**

_____ is not an operation of relational algebra.

- **Addition**
- Union
- Cartesian product
- Set difference

Candidate keys which are not selected as primary key are called _____

- Secondary keys
- Foreign keys
- **Alternate keys**
- Composite keys

Major benefits of data sharing is:

- Data of same applications placed at same place.
- Data of same applications placed at different places
- **Data of different application places at same place**
- Data of different applications stored at placed place

Functional dependency between two attributes or sets of attributes i.e. Q and P is represented as _____.

- **Q → P**
- Q P
- Q * P
- Q + P

Which of the following is not the component of Relational Data Model?

- Integrity Constraint
- **Manipulation Language**
- Relation / Table
- Domain Constraint

The attribute that is calculated from other attribute is called _____.

- **Derived attribute**
- Multi-valued attribute
- Composite attribute
- Simple attribute

Consider the two attributes “CNIC_NO” and “Registration_NO”. Both the attributes can be used for the unique identification. Now if the user selects the “CNIC_NO” as primary key, then “Registration_NO” is a ____ key.

- Secondary
- Primary
- **Alternative**

Which of the following constraint is included in the definition of the table?

- Default value
- Foreign key
- **Null**
- o Domain

The database deals with large amount of data. This property is called

- Database design
- **Scalability**
- Concurrency
- Robustness

Which one of the following is NOT a characteristic of meta data?

- Data about data
- Describes a data dictionary
- Self-describing
- **Includes user data**

Theta join is denoted mathematically as _____.

- R X S
- R X
- **R X S**
- RX RS

In cross reference matrix transitions are specified on _____ and database objects are specified on _____.

- Y axis, X axis
- X axis, Y axis
- X axis, X axis
- **Y axis, Y axis**

Each component of DBMS performs _____ functions.

- Similar
- **Different**
- Identical
- Computational

Which of the following data model is not record based?

- Hierarchical Data Model
- Network Data model
- **Semantic Data model**

The basic purpose of diamond in ER diagram is to show the _____.

- Attributes
- **Relationships**
- Composite attribute
- Entity set

Which of the following tool is used to design conceptual database design?

- SQL
- MS Access
- Oracle
- **This model is independent of any tool**

In “one to many” cardinality one instance of a relation is mapped with _____.

- **Many instances of second entity type**
- o One instance of second entity type
- Primary key of second relation
- Foreign key of second relation

Cross reference matrix is developed at the _____ stage of the database.

- Analysis
- **Designing**
- Implementation
- Application

The population of the data of the organization for which the database is created is called the

- _____ of the database.
- **Extension**
- Intension
- Model
- Schema

User rights information is stored in

- Physical database
- **Catalog**
- Logical database
- Buffer

Logical data independency provides independency to change _____ model.

- External
- Logical
- **Conceptual**
- Internal

In natural join the common attribute in the output table appears _____.

- Three times
- Four times
- **Once**

Which levels are mostly used for detailed DFD?

- **Level-0, Level-1**
- Level-1, Level-2
- Level-2, Level-3
- Level-3, Level-4

By default, a non-key attribute in a relation can have _____ value.

- Binary
- **Null**
- Character
- Integer

The three levels architecture is useful for _____.

- Arranging the data in an organized manner
- Hiding the details of internal systems
- Breaking the data access restriction
- Storing definitions of the structures

With the help of _____ technique data values with the smaller sized codes can further

- reduce the space needed by the data for storage in the database.
- **Compression**
- Expansions
- Negation
- Extension

If an entity is linked with itself, than it is called _____ relationship.

- Binary
- Ternary
- Nary
- **Recursive**

Duplication of data in controlled redundancy is controlled and _____.

- **Deliberate**
- Necessary
- Precise
- Unintentional

Teleprocessing is a sub-type of _____.

- Single user database
- **Multi-user database**

- Client server

In an ERD, the focus is on the _____ and the relationships between them.

- Attributes
- **Entities**
- Keys
- Indexes

The most widely used conceptual model is the _____ model.

- Implementation
- **ER**
- Object oriented
- Internal

Student and book are two entities, what is the cardinality between the two?

- Many to one
- One to many
- **Many to many**
- One to one

RDM is a simple there is just one structure and that is _____.

- **A relation or a table**
- E-R model
- Attributes
- Data model

A description on a particular collection of data using the given data model.

- **Database**
  - Relation
  - o Schema
- None

Consider thy key “CNIC, Name” and choose the best option that describes the key most effectively.

- The given key is a super key and alternate key.
- The given key is not a super key but a candidate key.
- **The given key is a super key and candidate key as well.**
- The given key is a super key but not a candidate key.

Normalization is a process of restricting a relation to

- **Minimize duplication of data in a database**
- Maximize duplication of data to ensure reliability
- Make if of uniform size
- Allow addition of data

Unary relationship involves a single entity it is also called _____.

- One to one relationship

- Many to many relationship
- **Recursive relationship**

Which of the following is a part of a DFD but not included in the ERD?

- The attribute having a single instance
- **The external entity having a single instance**
- The relationship having a single instance
- The role having a single instance

The database management system (DBMS) is used to

- Store the data
- Access the data
- Manage its users
- **All of above**

In cross reference matrix attributes are mention on _____ and reports are mention on

- _____.
- Y axis, X axis
- Y axis, Z axis
- **X axis, Y axis**
- X axis, Z axis

A table can be logically connected to another table by defining a _____.

- Hyper link
- Common attribute
- **Primary key**
- Super key

Data Redundancy means

- Security of data
- **Duplication of data**
- Management of data
- Recovery of data

Which of the following is a type of dependency?

- DBMS
- ERD
- DFD
- **Referential**

In your opinion, why relational database is widely acceptable?

- Due to its complexity
- Due to its approach
- **Due to its dependencies**

_____ is the first comprehensive complete database design.

- Logical database design
- Relational database design
- **Conceptual database design**
- Hierarchical database design

How data security can be assured in the database?

- By hiding data from all
- Through concurrent access of users
- By asking users not to share things
- **Implementing encryption algorithms**

The attributes whose value is not stored in a database is known as _____ attribute.

- Stored
- Single value
- Multi value
- **Derived**

Logical database design, like conceptual database design is our _____

- Data model
- Weak entity set
- Strong entity set
- **Database design**

If a single instance of "student" is associated with the many instances of the "subjects" entity, then the relationship between them is:

- One to one
- **One to many**
- Many to many
- Many to one

The conceptual database design is drawn through _____.

- Data model
- **ER model**
- Database design
- Entity set

A relational database is

- One that consist of two or more tables
- **A database that is able to process tables, queries, forms, reports and macros**
- o One that consists of two or more tables that are joined in some way
- The same as a flat file database

Database is a computerized representation of any organization's flow of _____ and storage of

- Data, information
- Details, data
- Data, details
- **Information, data**

In which stage of the database development process logical design is created?

- Requirement analysis
- Database design
- Physical design
- **Implementation**

Which of the following is NOT an entity?

- Employee
- Hobby
- Student
- **Playground**

After moving the determined columns from the original table to the new table while converting to 2NF, the determinate becomes the _____ of the new table.

- **Primary key**
- Foreign key
- Secondary key
- Composite key

The select operation works horizontally on the table, on the other hand the _____

- operates on a single table vertically.
- Binary operator
- **Project operator**
- Join operator
- Select operator

Free standing data dictionary is created by

- DFD
- Database
- **CASE tools**
- Cross reference matrix

A patient can be an outdoor or indoor. Analyze this statement and tell what type of

- constraint will be applied on patient?
- **Total completeness**
- Partial completeness
- Total disjoint
- Partial disjoint

If A → B and A → C, then A → BC. The inference Rule applies is:

- Decomposition
    - o Augmentation
- Reflexivity

Which of the following is the correct abbreviation of ERD?

- **Entity relationship diagram**
- Entity relation diagram
- Entity reality diagram
- Entity real distance

Database relation is also represented in a two dimensional structure called ____

- Row
- **Table**
- Entity
- Column

Data about data is metadata

- **True**
- False

Which of the following describes the job of a database administrator?

- Development, implementation operation of the physical database
- Creation of the system catalog
- **Monitoring and controlling database security and authorization: setting up controls to ensure the quality and integrity of data**
    - All of the above

Each table must have a _____ key.

- **Primary**
- Secondary
- Logical
- Foreign

A relational database system is based on the concept(s) of:

- A network of data records
- Nested tables
- **Tables, row and columns**
- A tree-like structure of data

A → BC and A → B, then A → C follow _____ inference rule.

- Augmentation
- Transitivity
- **Projectivity**
- Additivity

Suppose, we have two entities “shape” and “triangle”, then which of the following is a correct statement?

- Shape is a sub type entity and triangle is a super type entity
- Shape and triangle both are super type entities
- Shape and triangle both are sub type entities

The entity that is NOT dependent on any other entity for its existence is known as

- _____ entity.
- o Weak
- **Strong**
- Derived
- Dependent

In cross reference matrix attributes are specified on _____

- **X axis**
- Y axis
- Z axis
- None of the above

Which of the following clarify the semantics of a relationship?

- Entity
- **Roles**
- Relationship set
- Participants

How many types of cardinality is specified about relationship?

- Two
- One
- Three
- **Four**

Which of the following is NOT a component of a DFD?

- Dataflow
- Datastore
- External entities
- **Relationship between external entities**

Which of the following provides us the facility to add properties of one entity to another entity automatically?

- Abstraction
- Encapsulation
- **Inheritance**
- Dependencies

If there is more than one relationship between two entities and we mentioned role on the relationship link, then it is called:

- Unary relationship

- Ternary relationships
- **Multiple relationships**

Which one of the following is not an advantage of DBMS

- Data consistency
- **Recovery procedure is difficult**
- Faster development of new applications
- Better backup and recovery procedures

In many to many relationship a third table is created for the relationship, which is also called as _____

- Binary relation
- o **Associative entity type**
- Unary relationship
- Composite attributes

When one entity instance of another entity for its existence, then it is called _____

- **Existence dependency**
- Identifier dependency
- Referential dependency
- Acceptance dependency

Relational data model is widely accepted due to its mathematically proven foundation and its _____

- Complexity
- **Simplicity**
- Hierarchical structure
- Traditional structure

If an entity “motor cycle” is associated with at most one “driver”, then the relationship between these two entities is:

- **One to one**
- One to many
- Many to many
- Many to one

Database application is a program which is used for performing certain operations such

- as
- Insertion of data
- Extraction of data
- Updating the data
- **All of above**

Creating the relationships between new tables and their predecessors through the use of foreign keys is general requirements of _____

- First normal form

- Third normal form
- Fourth normal form

In _____ the domains of attributes of a relation are atomic, that is they consist of single units that cannot be broken down further.

- **First normal form**
- Second normal form
- Third normal form
- Augmentation

Atomicity is a feature of _____.

- **1NF**
- 2NF
- 3NF
- BCNF

The _____ can see entire information structure of the database.

- o **BA**
- DBTG
- Database user
- External viewer

The attribute whose value is not stored in a database is known as _____ attribute.

- Stored
- Single value
- Multi value
- **Derived**

_____ are the structure defined for placing data in the attributes while designing physical design of database.

- Objects
- Variables
- Attributes
- **Data types**

Suppose a relation has five columns and ten rows, select the correct option to identify cardinality of relation.

- Five
- **Ten**
- Fifteen
- Twenty

Which feature of database provides conversion from inconsistent state of DB to a consistent state ensuring minimum data loss?

- User accessible catalog
- Data processing

- Authorization service

The maximum PL/SQL size of data type “VARCHAR2” in DBMS is _____ bytes

- 42767
- 52767
- **32767**
- 28767

_____ is state of database where a “course” cannot be inserted in the table, because this course has not been registered to any “student”.

- **Insertion anomaly**
- Deletion anomaly
- Updation anomaly
- Redundancy

The maximum PL/SQL size of data type “BLOG” in DBMS is_____

- **4 gigabytes**
- 8 gigabytes
- 10 gigabytes
- 12 gigabytes

_____ states that in a relation no attributes of a primary key (PK) can have a null value.

- Referential integrity constraint
- Null constraint
- Domain constraint
- **Entity integrity constraint**

A database state where deletion of the information about the student record deletes the course information as well is called _____.

- Insertion anomaly
- Updation anomaly
- Redundancy
- **Deletion anomaly**

An entity type is

- Defined when the database is actually constructed
- A specific type such as an integer, text, date, logical etc.
- A coherent set of similar objects that we want to store data on (e.g. STUDENTS, COURSE, CAR)
- Defined by the database designer

“select” and “project” are the examples of _____

- **Unary operation**
- Binary operation
- Ternary operation

External scheme evolves as user needs are _____ over the time.

- Ends
- Modified
- **Switch**
- Stable

An attribute that is the collection of some other attribute(s) is known as _____

- attributes.
- Multi-valued
- Single-valued
- No-valued
- **Composite**

Which of the following is correct regarding Dataflow diagram?

- Single DFD is required to represent a system
- The dataflow must be bidirectional
- **Created at increasing levels of detail**
- Used to represent the relationships among the external entities

_____ are those attributes that are a combination of two or more than two attributes.

- Default value
- Null value
- Primary key
- o **Composite attributes**

Controlling redundancy in a database management system DOES NOT help to

- avoid duplication
- avoid unnecessary wastage of storage space
- **avoid unauthorized access to data**
- avoid inconsistency among data

Which of the following concepts is applicable with respect to 3NF?

- Full functional dependency
- Any kind of dependency
- **Transitive dependency**
- Partial functional dependency

Identify the operation which is NOT one of the parts of the five basic set operations in relational algebra?

- **Join**
- Union
- Cartesian Product
- Set Difference

- Physical data independence.
- Concurrency control.
- **Logical data independence.**
- Functional dependency

For a third normal form we concentrate on relations with one _____ key, and we eliminate transitive dependencies

- **Candidate**
- Composite
- Super key
- Primary key

The entity relationship diagram is used to graphically represent the _____ database

- model.
- Condensed
- Physical
- Logical
- **Conceptual**

The _____ determines that the link between two entities is optional or mandatory.

- Maximum cardinality
- **Minimum cardinality**
- Maximum degree
- Minimum degree

Select the most appropriate statement. Which of the following data model is used to design logical database design?

- **Relational**
- Hierarchal
- o bject oriented
- Any model can be used

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